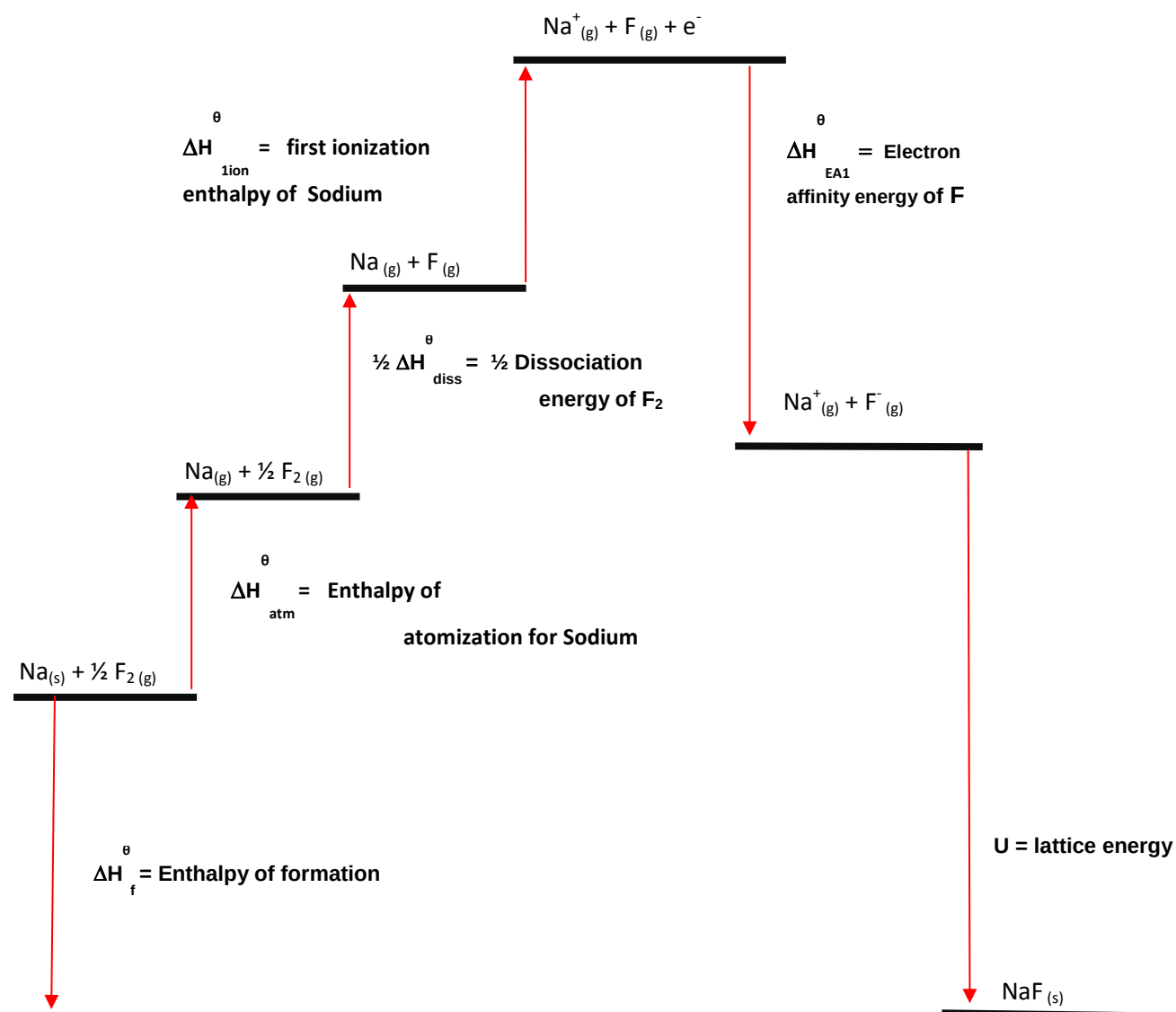


EXAM

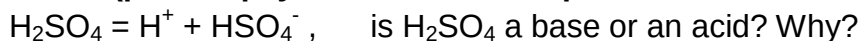
Inorganic Chemistry

- 1) Construct the Born-Haber cycle for sodium fluoride, NaF (Enthalpy numerical values can be omitted, 8pt).

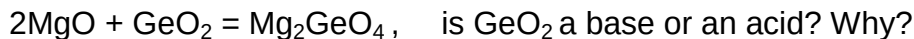


Please, remember to write down what type of enthalpy (atomization, dissociation, ...) is associated to each transition.

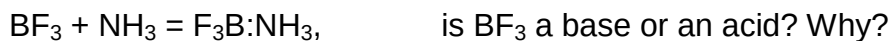
2) Use the most appropriate definition to identify acids and bases in the following reactions (please, pay attention to the specific reaction environment, 10pt)



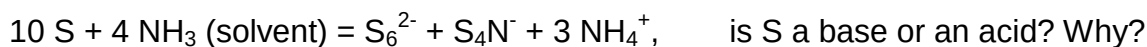
H_2SO_4 is an acid according to the Brønsted definition



GeO_2 is an acid according to the Lux-Flood definition



BF_3 is an acid according to the Lewis definition



S is an acid according to the "solvent system" definition



Co^{3+} is an acid according to the Lewis definition

3) Determine the oxidation state and the coordination number of the central atom for each of the following complexes (6pt):

Suggests possible geometries (linear, cubic, octahedral, ...) for these three compounds.

1) $(\text{NH}_4)_2[\text{Ni}(\text{C}_2\text{O}_4)_2(\text{H}_2\text{O})_2]$ $\text{C}_2\text{O}_4^{2-} = \text{oxalate}$
 $\hookrightarrow 2\text{NH}_4^+ + [\text{Ni}(\text{C}_2\text{O}_4)_2(\text{H}_2\text{O})_2]^{2-}$ Ligands: $2 \cdot \text{H}_2\text{O}$

$$x + 2 \times (-2) + 2 \times (0) = -2$$
 $2 \cdot \text{C}_2\text{O}_4 = (\text{bidentate})$
 $x = +2$ Coordination number: 6
 oxidation number: +2 geometry: octahedral

2) $[\text{Ni}(\text{NH}_3)_4]\text{Cl}_2$ Ligands: 4 NH_3
 $\hookrightarrow [\text{Ni}(\text{NH}_3)_4]^{2+} + 2\text{Cl}^-$ Coordination number: 4

$$x + 4 \times (0) = +2$$
 Geometry: { tetrahedral or
 $x = +2$ square planar }
 oxidation number: +2

3) $[\text{Fe}(\text{OH})(\text{H}_2\text{O})_5]^{2+}$ Ligands: 1 OH^-

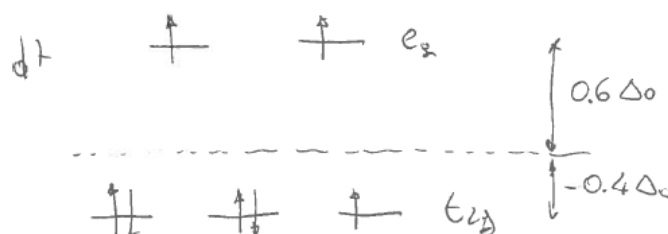
$$x + 1(-1) + 5 \times (0) = +2$$
 5 H_2O
 $x = +3$ Coordination number: 6
 oxidation number: +3 2 geometry: octahedral

4) Co_3O_4 belongs to the spinel-family. What is(are) the oxidation state(s) of cobalt in this mineral (2pt)? By using the crystal field theory, draw the energy configuration for the electrons in the d orbitals of the cobalt ions in this mineral, calculate the crystal field stabilization energy (CFSE) and try to predict if Co_3O_4 has normal spinel structure or inverse spinel structure (7pt).

The Spinel Family groups materials with the general formula AB_2O_4 , with A^{2+} and B^{3+} ,

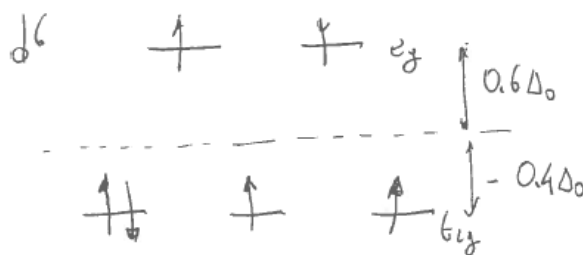
thus Co_3O_4 can be written as $(\text{Co}^{2+})(\text{Co}^{3+})_2(\text{O}^{2-})_4$ and the oxidation number for cobalt is both 2+ and 3+.

The electronic configuration for Co^{2+} is $[\text{Ar}] 3d^7$ and oxygen generally forms high spin complexes. Thus, we can draw the following diagram:



We can calculate an octahedral CFSE (crystal field stabilization energy) of $-0.8 \Delta_o$

The electronic configuration for Co^{3+} is $[\text{Ar}] 3d^6$ and oxygen generally forms high spin complexes. Thus, we can draw the following diagram:

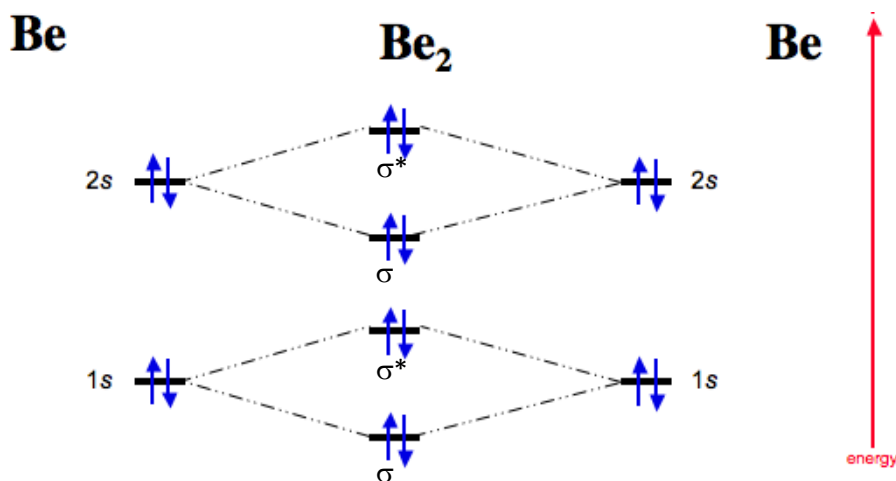


We can calculate an octahedral CFSE (crystal field stabilization energy) of $-0.4 \Delta_o$

This simple calculation suggests that a preferential occupation of the octahedral sites by the Co^{2+} ions and thus an inverse spinel structure.

(A consideration outside this exercise: this is one of the few cases where the Crystal Field Theory fails in predicting a spinel structure. In reality Co_3O_4 has normal spinel structure, because the stabilization field in the tetrahedral sites should also be taken into consideration, and because the $d^6 \text{Co}^{3+}$ ions prefer the low spin configuration)

5) Use the MO theory to predict the bond order in a Be₂ molecule. Comment the result (7pt).



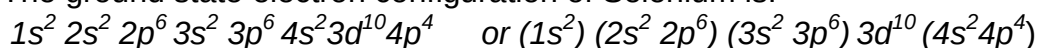
The bond order is $(4-4)/2=0$ or $(2-2)/2=0$.

Thus, thanks to the molecular orbital theory we can predict that the Be₂ molecule is unstable.

6) Using Slater's rules, calculate a value for the effective nuclear charge felt by an electron in a 3d orbital of a neutral Selenium (Se) atom (3pt). What are the limitations of the Slater's approach (1pt)?

The atomic number of Se is $Z = 34$

The ground state electron configuration of Selenium is:



In our textbook, the Slater's rules are enunciated as follow:

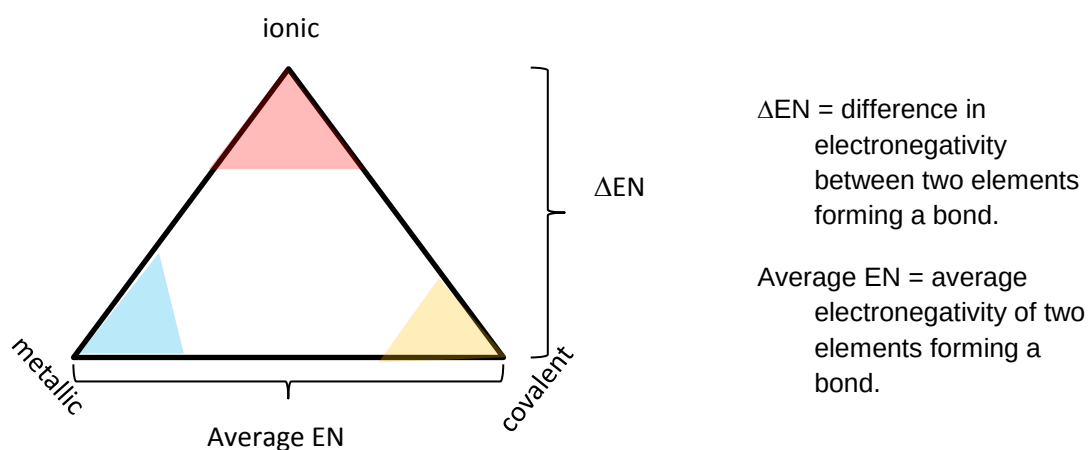
1. All electrons in orbitals of greater principal quantum number contribute zero.
2. Each electron in the same principal quantum number contributes 0.35, except when the electron studied is in a d or an f orbital; then the electrons in the s and p orbitals count 1.00 each.
3. Electrons in the $(n - 1)$ principal quantum level contribute 0.85 each, except when the electron studied is in a d or f orbital; then the electrons count 1.00 each.
4. All electrons in the lesser principal quantum levels count 1.00 each.

Hence, the shielding constant σ can be calculated as follow:

$$\sigma = (6 \times 0) + (9 \times 0.35) + (8 \times 1) + (8 \times 1) + (2 \times 1) = 21.15$$

$$Z^* = Z - \sigma = 34 - 21.15 = 12.85$$

3. Where can we find ionic, metallic and covalent bonds in the bond triangle? Explain your answer (2pt.). Describe the differences between ionic solids and covalent networks (2pt.).



If two elements have a large difference in electronegativity, the one with the highest electronegativity has the tendency to attract one or more electrons from the least electronegative element, thus forming an anion and a cation and creating a ionic compound.

If both the elements have low electronegativity, they will “like” sharing electrons and thus they will form a metallic alloy.

If both elements have high electronegativity, they have the tendency to form covalent bonds, i.e. to share localized electrons, and will form covalent molecules or covalent networks.

Ionic compounds are “keep together” mainly by electrostatic interactions, while covalent networks are tridimensional structures held together by covalent bonds.

4. What is the Madelung constant (2pt.)? Can we apply it to different crystals (1pt.)?

The Madelung constant is used in determining the electrostatic potential of a single ion in a crystal structure and accounts for the reciprocal position of the ions in the crystal lattice. Hence, different crystals will have the same Madelung constant only if they have the same lattice structure.